

	Books and References 1. S.C. Gupta and V. K. Kapoor, Fundamentals of Mathematical Statistics, Sultan Chand and Sons 2. Mood, A.M. Graybill, F.A. and Boes, D.C. (2007): Introduction to the Theory of Statistics, 3rd Edn., (Reprint), Tata McGraw-Hill Pub. Co. Ltd. 3. Christian Heumann, Michael Schomaker and Shalabh (2016): Introduction to Statistics and Data Analysis with Exercises, Solutions and Applications in R., Springer International Publishing Switzerland 4. John E Freund (2014): Mathematical Statistics, Pearson Edn, New Delhi 5. Rohatgi V. K. and Saleh, A.K. Md. E. (2009): An Introduction to Probability and Statistics. 2ndEdn. (Reprint) John Wiley and Sons.		

SEMESTER IV

Programme	B. Sc. Statistics				
Course Code	STA4CJ201(P)				
Course Title	Probability Distributions				
Type of Course	Major				
Semester	IV				
Academic Level	200-299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	HSE level Mathematics/Statistics courses				
Course Summary Objective	To understand random variables, their probability distributions (discrete and continuous cases separately). To analyse their characterization & properties of the distribution. To gain proficiency in transformation of random variables. To analyse their characterization & properties of the real data set.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Apply probability concepts to model.	U	C	Instructor-created exams / Quiz
CO2	Analyze random phenomena Analyze the corresponding distribution and its characterization.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Gain thorough idea about theoretical and practical aspects of Probability distribution	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Apply transformation of random variables to obtain new distributions.	U	C	Instructor-created exams / Home Assignments
CO5	Uses of moments, cumulates, and characteristic functions.	Ap	P	One Minute Reflection Writing assignments
CO6	Analyze the distributional properties of data using moments, skewness, and kurtosis.	Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Standard Distributions Discrete		10	16
	1	Degenerate distribution		
	2	Bernoulli distribution (definition).		
	3	Binomial distribution (definition, properties and application).		
	4	Poisson distribution (definition, properties and application).		
	5	Relationship between Binomial and Poisson Distributions		
	Sections from References:			
II	Standard Distributions Discrete		10	16
	6	Discrete Uniform distribution (definition and basic properties).		
	7	Geometric distribution (definition and basic properties).		
	8	Lack of memory property of Geometric Distribution		

	9	Negative Binomial distribution (definition and basic properties).		
	10	Hyper-geometric distribution (definition and basic properties).		
	Sections from References:			
III	Standard distributions Continuous		10	18
	9	Rectangular (definition, mean, variance and mgf)		
	10	Exponential (definition, mean, variance and mgf)		
	11	Memoryless property of Exponential distribution		
	12	Gamma (definition, mean, variance and mgf)		
	13	Beta (definition, mean, variance and mgf)		
	14	Relationship between Gamma, Beta first and second distributions		
	Sections from References:			
IV	Normal distribution		15	20
	15	Definition		
	16	Derivation of Mean and Variance		
	17	Derivation of Median and Mode Mean Deviation		
	18	Derivation of MGF		
	19	Derivation of General Central Moment		
	20	Standard Normal Distribution		
	21	Normal distribution- additive property		
	22	Area properties of Normal Distribution		
	24	Computing area under standard Normal Curve		
	25	Quartile Deviation		
	26	Lognormal, Pareto Distributions (definition only).		
	27	Cauchy, Weibull and Laplace Distributions (definition only).		
	Sections from References:			
V	Open Ended Module: Practical Applications		30	
	1	Fitting of standard distributions Random number generation using software		
	Sections from References:			
Books and References:				
1. S.C. Gupta and V.K. Kapoor : Fundamentals of Mathematical Statistics, Sultan Chand and sons				
2. V.K. Rohatgi: An introduction to Probability theory and Mathematical Statistics, Wiley Eastern.				
3. Mood A.M., Graybill. F.A and Boes D.C. : Introduction to Theory of Statistics McGraw Hill				
4. Johnson, N.L., Kemp, A.W., and Kotz, S.. (2005): Univariate Discrete Distributions, 5th edition, Wiley Interscience, John Wiley & Sons				
5. Johnson, N.L., Kotz, S., and Balakrishnan, N. (2002): Continuous Univariate Distributions, Vol. 1, John Wiley				
6. Johnson, N.L., Kotz, S., and Balakrishnan, N. (2002): Continuous Univariate Distributions, Vol. 2, John Wiley				
7. Hogg, R. V., Craig, A., and Mckean, J.W. (2019): Introduction to Mathematical STATistics, 8th edition , Pearson				
8. John E Freund : Mathematical Statistics (Sixth Edition), Pearson Education (India),New Delhi.				